The Future of Software Engineering

Software Engineering

- It was about problem solving
- It is about problem solving
- It will be about problem solving (with AI)
    - The software engineers with AI skills will solve problems 10 or 100 times faster than those without AI skills.

Learing how to solve problems

- We should learn how to solve problems with any tools available (classroom meetings, reading books, or watching online lectures), including AI.
- The best way to learn how to solve problems is to solve problems, and learn from the process.
- In other words, make something and fail, most great problems solvers learn this way.

The miracle - AI

- We can learn much faster and better with AI, and it knows (seems to know) everything, so we can ask it anything.
- It can help us solve problems, and it can also help us learn how to solve problems much faster and better.
- Only the software engineers who can use this revolutionary tool will survive.

Arguments about AI

Current (state of the art) AI is about solving specific problems, but we are waiting for the GenAI that solves problems just like human beings.

- Some say it will arrive in 2029 (Ray Kurzweil)
- Some say it will never arrive (Elon Musk).

No matter what the answer migh be, we know that:

1. The AI that we are using now is the worst AI that we will ever have.
2. The current AI is already too powerful to replace all the coders and programmers.
3. Fortunately, software engineers are problem solvers, so as long as we can solve problems (with or without AI), we will survive and thrive.

The future of software engineering

1. Just like we used local computers and web at the same time for different purposes, we will use local AI and cloud AI.
2. For simple coding and programming tasks, we will use local AI, and for more complex and large scale problems, we will use cloud AI.

3. The hallucination of AI will be reduced, and tools such as Ontology and Knowledge Graph will be used to improve the accuracy and reliability of AI (Palantir as an example).
4. The software engineers will be called as "System builder", "System architect", or "Problem Solver" in general.

5. Context engineering and Harness engineering will be some of the most important skills.
6. Software engineers need to use the 2nd brain software tool to organize their thoughts and ideas from/with the interaction of AI.
7. Software engineers will use old but new technologies, such as automation, CLI, and scripting languages, to solve problems faster and better with AI.

8. Understanding and building LLMs will be the next Python learning/coding, it will be the common sense.
9. Integrating all the tools (AI, software engineering, programming, thinking, and others) together to solve problems will be the most important skill, and it will be the most difficult skill to learn.

When we can do this, we will survive and thrive; if we can't, we have no hope.